

Math Primer

(12. Optimization II – Non-Linear Programming)

Contents



- ▶ Convex Set
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- ▶ Lagrangian Method
- ▶ KKT Conditions



JL Lagrange
(1736-1813)

-
- A diagram showing a blue, irregularly shaped region labeled S in red. Inside the region, two points are marked: x_1 (with a red underline) and x_2 (with a red underline). A line segment connects x_1 and x_2 . The region is also labeled $\mathcal{U}$ in red at the bottom left.

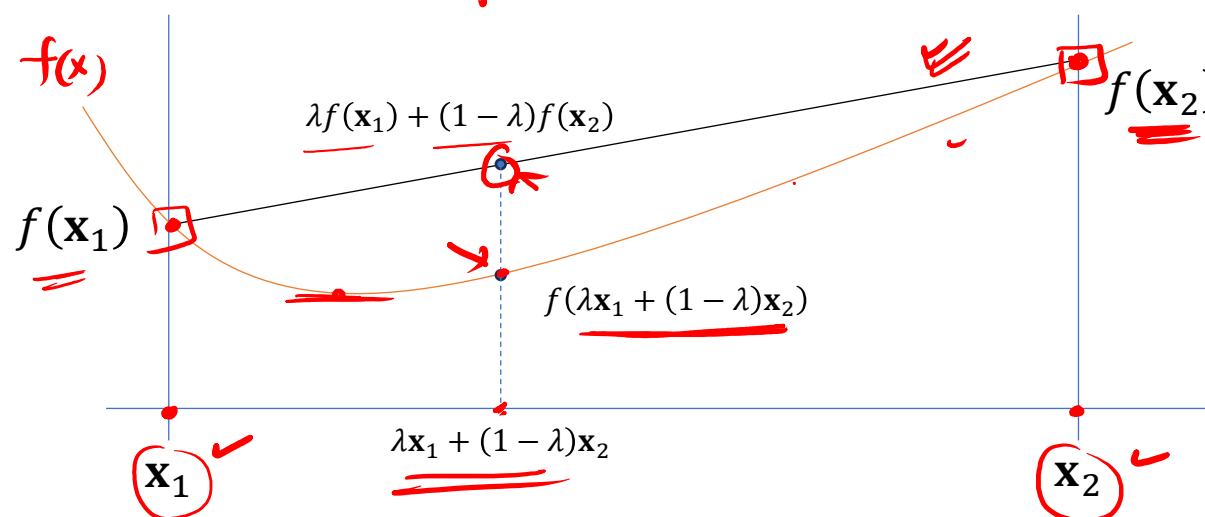
A diagram illustrating a set S (shaded orange) with two points x_1 and x_2 (circled in red). A line segment connects x_1 and x_2 , and a red squiggly line indicates a path or distance between them.

Mathematically, for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^n$, if $S \subseteq \mathbb{R}^n$ and $\lambda \in [0,1]$, then $\lambda \mathbf{x}_1 + (1 - \lambda) \mathbf{x}_2 \in S$

Convex Function

► If $S \subseteq \mathbb{R}^n$ is a convex set, then a function $f: S \rightarrow \mathbb{R}$ is said to be convex over S if

$$\lambda f(\mathbf{x}_1) + (1 - \lambda)f(\mathbf{x}_2) \geq f(\lambda \mathbf{x}_1 + (1 - \lambda)\mathbf{x}_2), \lambda \in [0, 1]$$



→ Convex Objective functions are preferred because they have global minima

A function is convex if the Hessian Matrix is positive semi-definite

$$\mathbf{z}^T \mathbf{H} \mathbf{z} \geq 0$$

$$\mathbf{z}^T \mathbf{H} \mathbf{z} > 0$$

Convex Function



Exercise : identify the following functions as Convex or Concave

- a) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 \rightarrow \text{Convex}$ 
- b) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x| \rightarrow \text{Convex}$ 
- c) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x \rightarrow \text{Convex}$ 
- d) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = -x^2 \rightarrow \text{Concave}$ 
- e) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \ln x \rightarrow \text{Concave}$ 
- f) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 3 \rightarrow \text{both}$
- g) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \sin x \rightarrow \text{neither}$ 

Convex Programming

- We call the following problem

$$2x_1 + 3x_2 \leq 5$$

$$\Rightarrow \underline{2x_1 + 3x_2 - 5 \leq 0} \quad (g_1)$$

$$g_2: \underline{x_1 - x_2 \leq 0}$$

$$5x_1 + 2x_2 \geq 3 \Rightarrow \underline{-5x_1 - 2x_2 + 3 \leq 0} \quad (g_3)$$

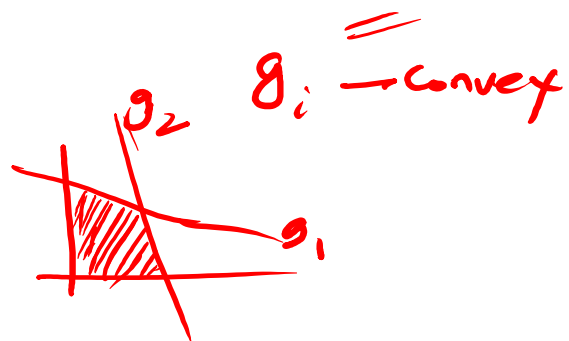
$$\begin{aligned} & \min f(\mathbf{x}), \text{ where } \mathbf{x} = (x_1, x_2, \dots, x_n) \\ & \text{s.t. } \underline{g_i(\mathbf{x}) \leq 0}, \text{ where } i = \underline{1, 2, \dots, m} \end{aligned}$$

$\mathbf{x} \in \mathbb{R}^n$

max $\boxed{f}$ concave
min $\boxed{-f}$

Convex programming problem if f and g are **convex functions** in the above formulation. Note that the objective function is of the type 'minimize' and constraints are of the type ≤ 0 . If the specification is opposite, then we would need the corresponding functions as concave.

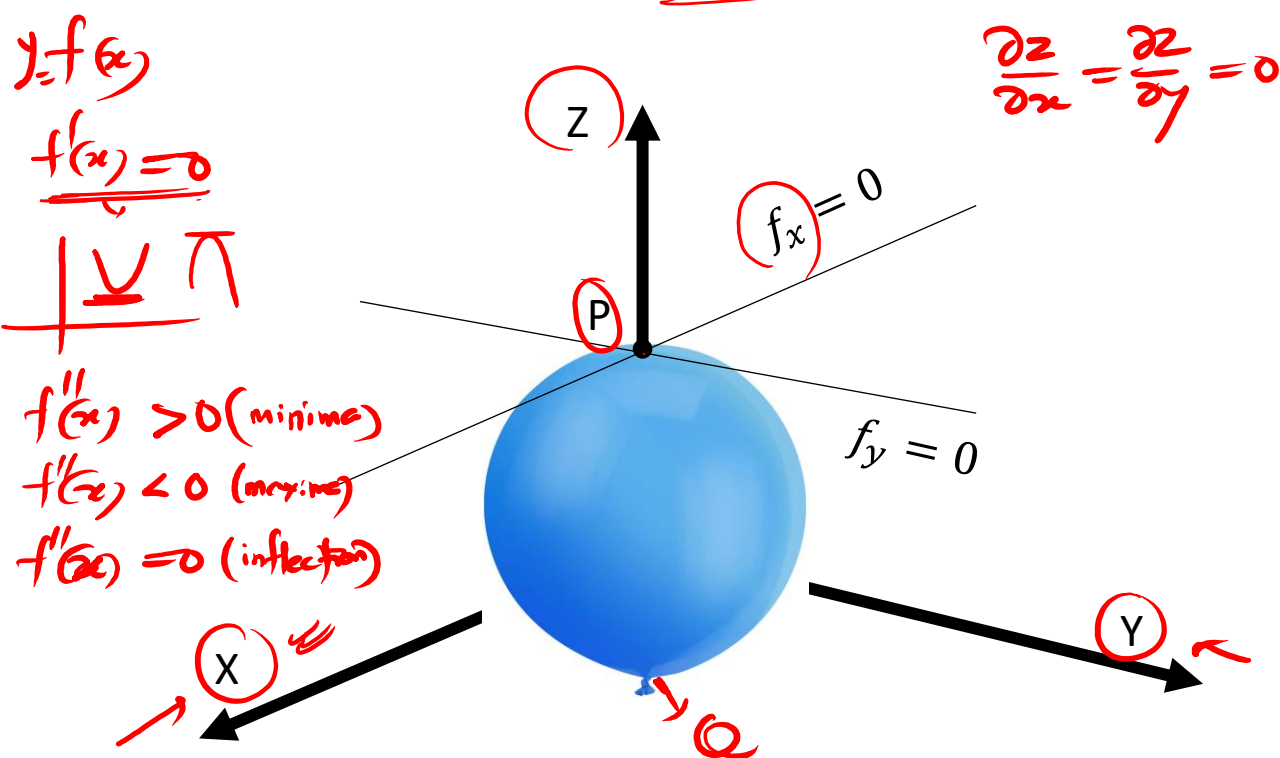
- The constraints being convex functions create a convex set



$$S = \{x \in \mathbb{R}^n; g_i(\mathbf{x}) \leq 0, i = 1, 2, \dots, m\}$$

Unconstrained Optimization

- Just like we had first order and second order conditions for local minima/maxima in univariate case, we do have similar conditions in multivariate case ; although it's a little complicated. Below is an example of a bivariate case



stationary points

Step - 1 : Find critical points by setting $f_x = 0$ and $f_y = 0$

Step - 2 : Find 2nd order partial derivatives at the critical points

f_{xx}

f_{yy}

f_{xy}

$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$

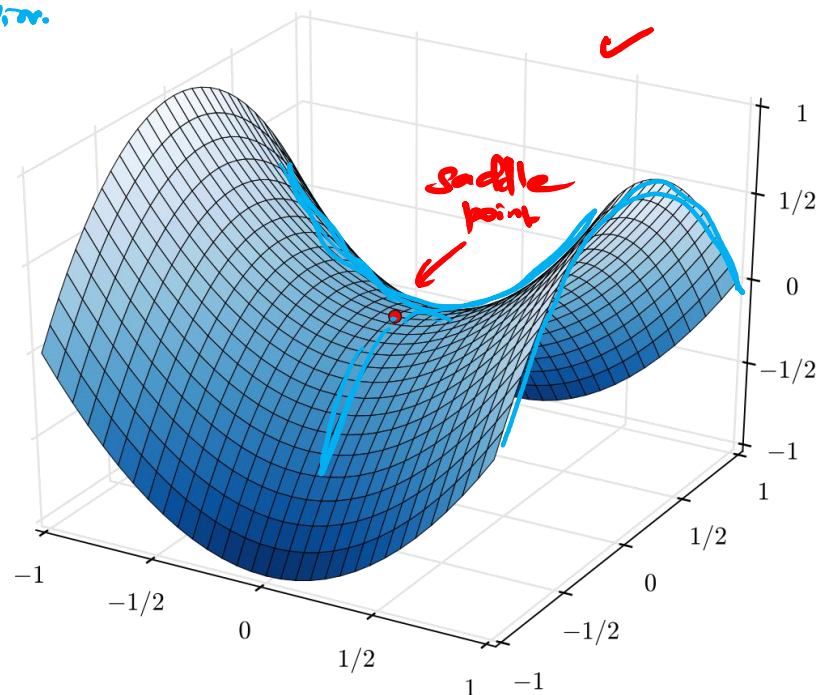
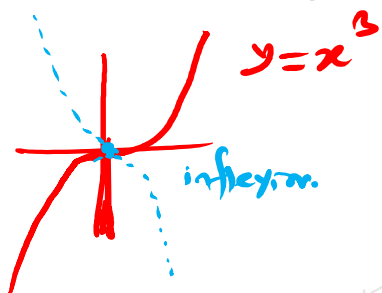
Step - 3 : Compute determinant of Hessian Matrix $D(x, y) = f_{xx}f_{yy} - f_{xy}^2$

Step - 4 :

1. If $D(x, y) > 0$ $f_{xx} > 0$ (minima) ✓
2. If $D(x, y) > 0$, $f_{xx} < 0$ (maxima) ✓
3. If $D(x, y) < 0$ (saddle point) ✓
4. If $D(x, y) = 0$ (inconclusive)

Saddle Point

► Saddle point is the equivalent of point of inflexion in higher dimensions



Conditions in Multivariate Settings

► The general conditions for optimality under multivariate settings are given as

→ First Order Condition : $\nabla f(\mathbf{x}) = \mathbf{0}$ (gives stationary points) $\begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Second Order Condition :

Case 1 : $\mathcal{H}_f$ (Hessian) is positive definite ^{>0} implies local minimum

Case 2 : $\mathcal{H}_f$ (Hessian) is negative definite ^{<0} implies local maximum

Case 3 : $\mathcal{H}_f$ (Hessian) has both positive and negative eigen values implies saddle point

Case 4 : Else the result is inconclusive

Exercise



► Find the stationary points of the following functions and comment if they are minima/maxima/saddle point/inconclusive

a) $f(x, y) = 3x^2 + 7xy + 5y^2$

b) $g(x, y) = 3x^2 + 8xy + 5y^2$

Constrained Optimization – Equality Constraints

- Now we will include constraints. Consider non-linear optimization problem with equality constraints. It can be solved using **Lagrangian Function**

$$\begin{aligned} \min f(\mathbf{x}) \text{ , where } \mathbf{x} &= (x_1, x_2, \dots, x_n) \\ \text{s.t. } g_i(\mathbf{x}) &= 0 \text{ , } i = 1, 2, \dots, m \end{aligned}$$

Handwritten notes: Blue arrows point from λ to $x_1, x_2, \dots, x_n$ and from λ_i to x_i . A blue circle around (x_j, λ_i) is labeled $(m+n)$.

We write the Lagrangian function as

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i (g_i(\mathbf{x}))$$

Handwritten notes: Blue arrows point from $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$ to "(Lagrangian Function)". Blue arrows point from λ_i to "(Lagrangian Multiplier)".

The constrained optimization is now converted to unconstrained optimization problem.



Non-Linear Optimization with Equality Constraints

- **Necessary conditions** (first order partial derivative w.r.t all x_j and λ_i equal to 0)

$$\underline{x_1^*, x_2^* \dots x_n^*, \lambda_1^*, \lambda_2^* \dots \lambda_m^*}$$

$$\rightarrow \frac{\partial L}{\partial x_j} = 0, j = 1, 2, \dots, n$$

$$\rightarrow \frac{\partial L}{\partial \lambda_i} = 0, i = 1, 2, \dots, m$$

- **Sufficient conditions** – Suppose $Z(\mathbf{x}) = \{z \in \mathbb{R}^n : z^T \nabla g(\mathbf{x}^*) = 0\}$ and $z^T \mathcal{H}(\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*))z > 0$ for all $z \in Z(\mathbf{x})$ and $z \neq 0$ then the point is local minima

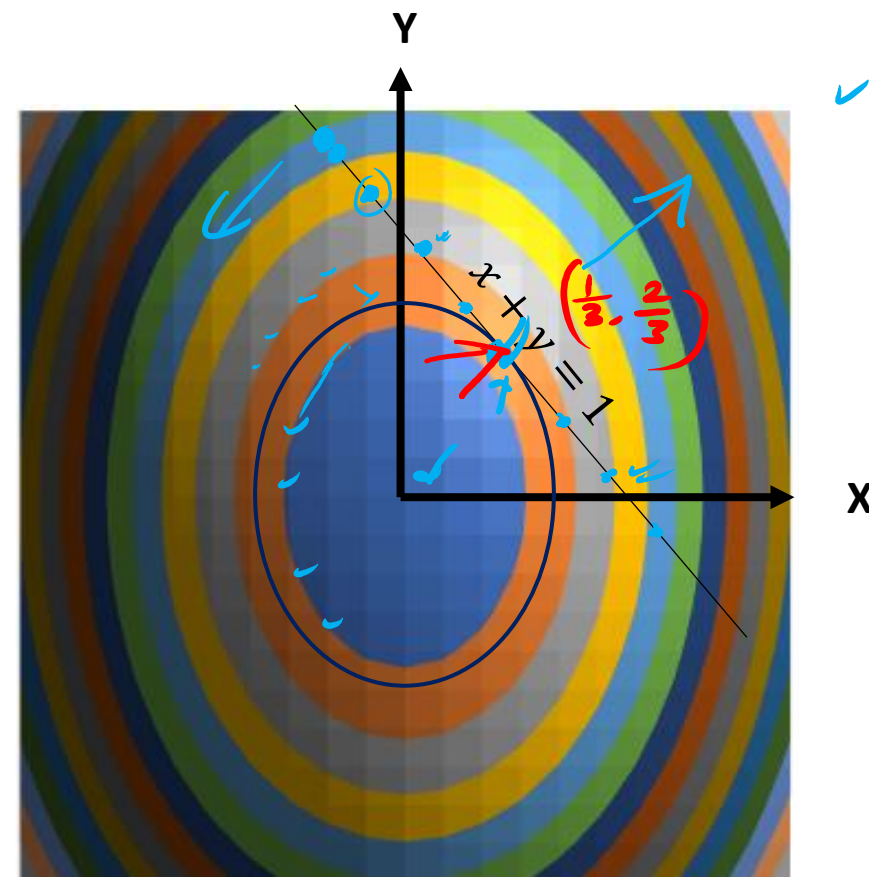
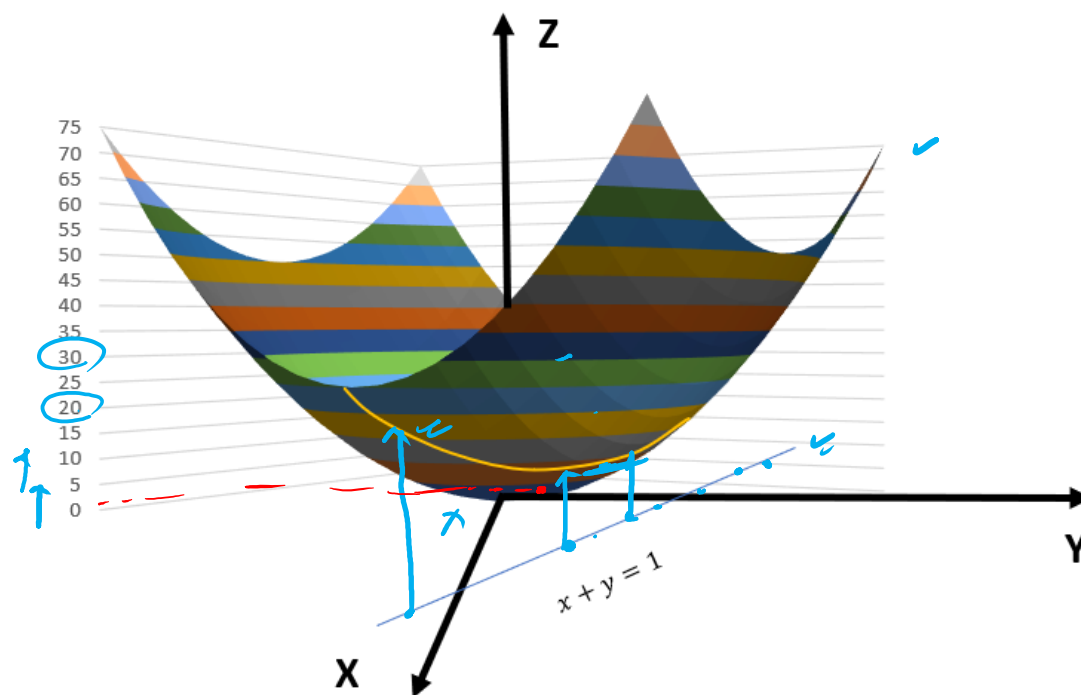
else if $z^T \mathcal{H}(\mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*))z < 0$ for all $z \in Z(\mathbf{x}^*)$ and $z \neq 0$ then the point is local maxima

Exercise - What is the highest possible product of 2 positive numbers given that the sum is 20?

More Examples with Visuals

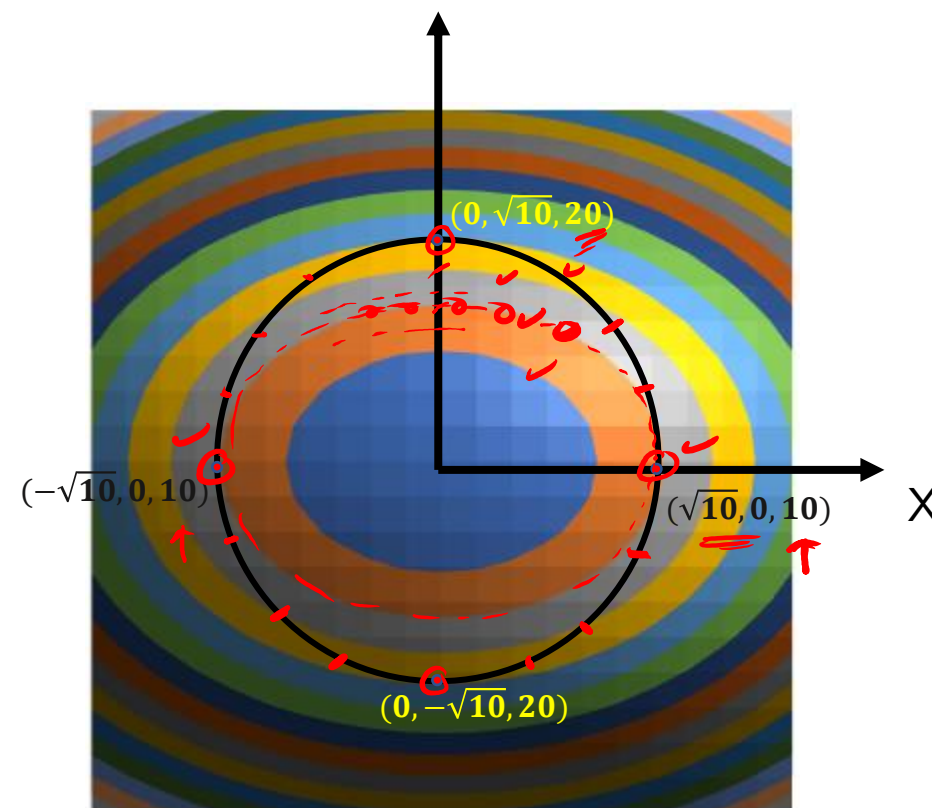
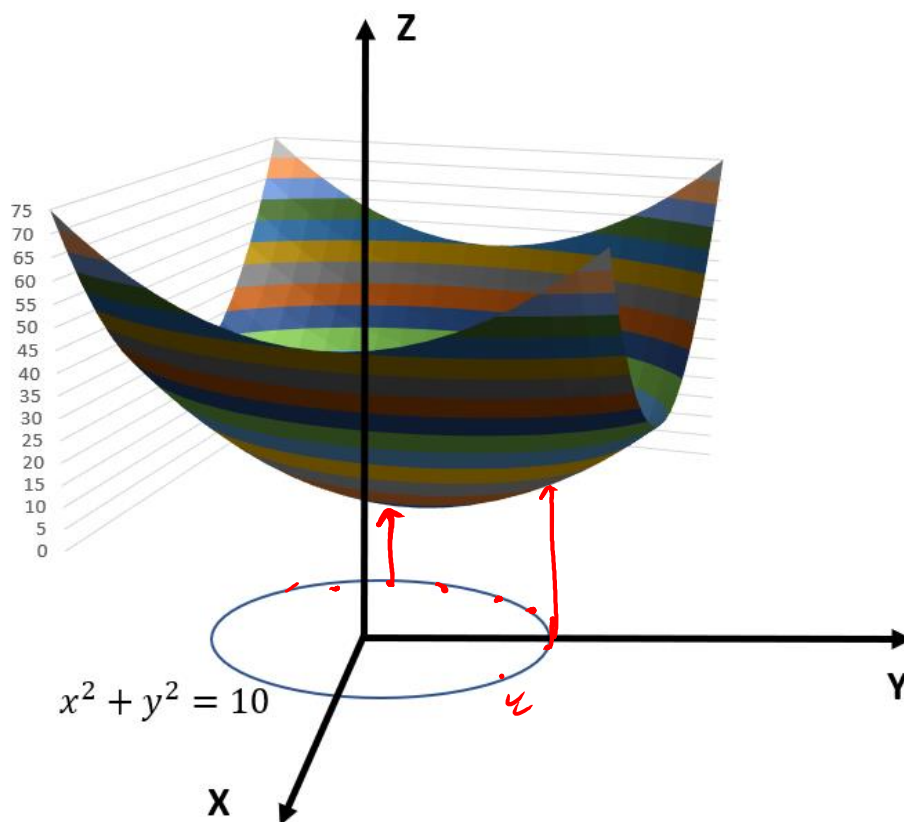
$$\min z = f(x, y) = 2x^2 + y^2$$

$$\text{s.t. } x + y = 1$$



More Examples with Visuals

$$\begin{array}{ll} \text{Min} & \underline{z = f(x, y) = x^2 + 2y^2} \\ \text{s.t} & \underline{g(x, y) = x^2 + y^2 = 10} \end{array}$$



Non-Linear Optimization with Inequality Constraints

- For non-linear optimization problems with inequality constraints, we need to understand **Karush-Kuhn-Tucker (KKT)** conditions. Consider the following constrained optimization problem with inequality constraints

$$\begin{aligned} \rightarrow \min f(\mathbf{x}) \text{ , where } \mathbf{x} &= (x_1, x_2, \dots, x_n) \\ \text{s.t. } g_i(\mathbf{x}) &\leq 0 \text{ , } i = 1, 2, \dots, m \end{aligned}$$

The necessary KKT conditions for optimality are

$$\rightarrow \textcircled{1} \quad \nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) = \mathbf{0}$$

$$\textcircled{2} \quad g_i(\mathbf{x}^*) \leq 0 \text{ , } i = 1, 2, \dots, m \quad \checkmark \checkmark$$

$$\textcircled{3} \quad \lambda_i^* g_i(\mathbf{x}^*) = 0 \text{ , } i = 1, 2, \dots, m \quad \checkmark$$

$$\textcircled{4} \quad \lambda_i^* \geq 0 \text{ , } i = 1, 2, \dots, m \quad \checkmark$$

$\lambda \quad g(\mathbf{x}^*) < 0$
 $\downarrow$
 (binding) $g(\mathbf{x}^*) = 0, \lambda \geq 0$
 (non-binding) $g(\mathbf{x}^*) < 0, \lambda = 0$

Non-Linear Optimization with Inequality Constraints



- **Sufficient Condition** – Let $(\mathbf{x}^*, \lambda^*)$ satisfy all 4 necessary conditions. If f and g are differentiable convex functions, then the solution is a global minima

Exercise :

1.
$$\begin{aligned} \min f(\mathbf{x}) &= 2x_1 + x_2 \\ \text{s.t. } x_1^2 + x_2^2 &\leq 4 \\ x_1 - x_2 &\leq 0 \end{aligned}$$

2.
$$\begin{aligned} \min f(\mathbf{x}) &= x_1^2 + x_2^2 - 2x_1 \\ \text{s.t. } x_1^2 + x_2 - 1 &\leq 0 \end{aligned}$$

Non-Linear Optimization with Inequality Constraints



Homework

$$\min -(x - 2)^2 - 2(y - 1)^2$$

$$\begin{aligned} \text{s.t.} \quad & x + 4y \leq 3 \\ & -x + y \leq 0 \end{aligned}$$